

Business Requirements Document (BRD)

Project Title: Indian Weather Forecasting & Data Analysis

Prepared By: Avadhoot Anil Jadhav.

Date: 11/12/2025

1. Project Overview :

This project focuses on the comprehensive analysis of Indian weather data, aiming to uncover statistical distributions, temporal variations, and key trends in meteorological parameters across various locations. The analysis integrates data from multiple sources concerning astronomical conditions, weather forecasts, and geographical locations to provide a holistic view of the weather patterns.

The ultimate goal of this analysis is to:

- Understand the typical range and distribution of key weather variables like temperature.
 - Identify hourly or daily patterns in dynamic variables such as wind speed.
 - Provide foundational insights for potential weather forecasting or environmental studies.
-

2. Objectives and Goals :

The key objectives of this analysis are:

- To perform **Distribution and Variation Analysis** of core weather parameters (e.g., Temperature, Wind Speed)¹.
 - To **visualize trends** and distributions through various graphs².
 - To determine the average temperature and the range in which most recorded temperatures fall³.
 - To identify **peak and low periods** for dynamic weather variables, such as the highest and lowest average wind speeds throughout the day⁴.
 - To study the relationship between time (hourly/daily) and changes in weather conditions⁵.
-

3. Data Description :

The analysis uses multiple datasets that were merged into a single comprehensive dataset. The core data sources were: astro_data.csv, forecast_data.csv, and location_data.csv⁶.

Column Name	Description
city	The name of the city/location ⁷ .
state_x	The state of the location ⁸ .
time_epoch	Epoch time stamp of the recording ⁹ .
time	Date and time of the weather recording ¹⁰ .
sunrise, sunset, moonrise, moonset	Astronomical timing data ¹¹ .
moon_phase, moon_illumination	Descriptive astronomical conditions ¹² .
dewpoint_f	Dewpoint temperature in Fahrenheit ¹³ .
will_it_rain	Indicator for the probability of rain ¹⁴ .
gust_mph, gust_kph	Wind gust speed in miles and kilometers per hour ¹⁵ .
Other	Various temperature, pressure, and humidity-related fields (implied from the analysis context).

4. Functional Requirements (Key Analysis Steps) :

a. Data Preprocessing

- Load the three primary datasets: astro_data.csv, forecast_data.csv, and location_data.csv¹⁶.
- Rename necessary columns (e.g., renaming the location column to city for consistency)¹⁷.

- Merge the datasets based on common identifiers (like city) to create a unified dataset¹⁸.

b. Exploratory Data Analysis (EDA)

- **Distribution Analysis:** Analyze the statistical distribution of meteorological parameters¹⁹.
 - **Temperature Variation Study:** Evaluate the average temperature and the interquartile range of recorded temperatures²⁰.
 - **Hourly Wind Speed Trend Analysis:** Study how wind speeds vary throughout the day²¹.
 - **Graph Visualization:** Generate appropriate graphs to illustrate key distributions and temporal trends²².
-

5. Key Insights and Deliverables :

The analysis yielded the following key findings:

- **Temperature:** The temperature readings in the dataset are around an average of about 25°C. Most of the recorded temperatures fall between 22.5°C and 28.1°C, though the dataset includes a minimum reading of 1.5°C.
 - **Wind Speed Trends:** Wind speeds are generally highest during the day, peaking around 12:00 PM with an average of about 10 kph. Wind speeds are typically lowest in the early morning (around 6:00 AM) averaging close to 7 kph.
 - **Deliverables:**
 - A detailed analysis report summarizing the statistical distributions and time-based trends.
 - Visualizations (graphs) illustrating the findings²⁸.
-

6.Tools and Technologies :

The analysis will be performed using the following tools:

- Programming Language: Python
- Libraries: Pandas, NumPy, Matplotlib, Seaborn.
- Development Environment: Jupyter Notebook